

Beyond teffects and lateffects: Average and Local Average Treatment Effects with Covariates in Stata

Tymon Słoczyński

Brandeis University

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Introduction

Stata provides `teffects` and `lateffects` for estimating **average and local average treatment effects with covariates**, respectively.

- `teffects` was introduced in Stata 13
- `lateffects` is available in StataNow 19

In this presentation, I will discuss a suite of related commands – `drlate`, `kappalate`, and `teffects2` – that I have developed together **with S. Derya Uysal and Jeffrey M. Wooldridge** as companion software for our three recent papers on treatment effect estimation.

- All three packages are available from SSC (`ssc install ...`).

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- All three packages are available from SSC (`ssc install ...`).

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- 2 Papers
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Paper #1

Śłoczyński, Uysal, and Wooldridge (2025). “Abadie’s Kappa and Weighting Estimators of the Local Average Treatment Effect.” *JBES* 43(1), 164–177.

- A unified treatment of **weighting estimators of the LATE**, including estimators motivated by **Abadie’s (2003) kappa theorem**.
- Shows that **normalization is substantively important**: unnormalized estimators are not invariant to recentering the outcome and, with logarithmic outcomes, can depend on the original measurement units.
- Identifies a **preferred normalized estimator** whose denominator is positive by construction under either form of one-sided noncompliance. The instrument propensity score can be estimated by **maximum likelihood or covariate balancing** (cf. Heiler 2022; Sant’Anna et al. 2022).
- Develops asymptotic theory and inference.
- Implemented in `kappalate`.

Paper #2

Słoczyński, Uysal, and Wooldridge (2022). “Doubly Robust Estimation of Local Average Treatment Effects Using Inverse Probability Weighted Regression Adjustment.” <https://arxiv.org/abs/2208.01300>.

- Introduces **inverse probability weighted regression adjustment (IP-WRA)** estimators of the LATE and LATT when covariates are used to make the instrument conditionally exogenous or to improve precision.
 - ▶ The Stata reference manual for `teffects` refers to this class of estimators as “**Wooldridge’s double-robust.**”
- The estimators are **doubly robust**: they remain consistent if either the instrument propensity score model or the relevant conditional mean models are correctly specified.
- Discusses improved finite-sample stability relative to AIPW estimators.
- Develops straightforward inference and a **doubly robust Hausman-type test of unconfoundedness** that remains valid with heterogeneous treatment effects (cf. Donald et al. 2014).
- Implemented in `drlate`.

Paper #3

Śłoczyński, Uysal, and Wooldridge (2026). “Covariate Balancing and the Equivalence of Weighting and Doubly Robust Estimators of Average Treatment Effects.” <https://arxiv.org/abs/2310.18563>.

- Shows that estimating propensity scores using **inverse probability tilting (IPT)**, as in Graham et al. (2012, 2016), produces **exact covariate balance and automatically normalized weights**.
- As a result, **IPT-based IPW, AIPW, and IPWRA estimators are numerically identical** for the ATE and ATT when the augmentation or regression adjustment models are linear.
- Thus, a seemingly consequential **choice among weighting and doubly robust estimators disappears** once the propensity score is estimated using the appropriate balancing conditions.
- Extends these equivalence results to IV estimators of the LATE and difference-in-differences estimators (cf. Sant’Anna and Zhao 2020).
- Implemented in `teffects2`.

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kappalate

Syntax: `kappalate depvar (treatment = instrument) [indepvars]
[if] [in] [, options]`

Essential options:

- `zmodel(string)` selects the **approach to estimating the instrument propensity score**; options include `logit`, `probit`, and `cbps`, where `cbps` uses the “covariate balancing propensity score (CBPS)” approach of Imai and Ratkovic (2014) and Heiler (2022)
- `which(string)`, where `string` may be `all` or `norm`, which determines whether all estimates or only normalized estimates are displayed

kappalate vs. lateffects

Many features of kappalate overlap with those of lateffects `kappa` and `lateffects` balancing.

Advantages of kappalate:

- A broader set of estimators, especially with option `which(all)`
- Allows nonbinary treatments
- Automatic standardization of nonbinary covariates
- Available for Stata 17, 18, and 19

Advantages of lateffects:

- Allows covariate balancing (CBPS) estimation with both the logit and probit models
- Works with weighted data
- Likely faster

Normalized kappa with lateffects

```
. quietly webuse card95  
  
. generate expersq = exper^2  
  
. lateffects kappa (lwage) (somecol) (nearc4 black south smsa smsa66 region ex  
> per expersq)
```

Iteration 0: EE criterion = 1.138e-21

Iteration 1: EE criterion = 9.837e-30

Local average treatment effect Number of obs = 3,010

Estimator: Normalized kappa

Outcome model: Weighted mean

Treatment model: Weighted mean

IV pscore model: Logit

	lwage	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]

LATE						
somecol						
(Yes vs No)		.3590843	.2271338	1.58	0.114	-.0860898 .8042584

Normalized estimators with kappalate

```
. kappalate lwage (somecol=nearc4) black south smsa smsa66 region exper expers  
> q, zmodel(logit)
```

Weighting estimation of the LATE

```
Outcome      :   lwage  
Treatment    :   somecol  
Instrument    :   nearc4  
IPS          :   logit ML
```

```
Number of obs   =   3010
```

lwage	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----						
tau_a,10	.3296137	.2300512	1.43	0.152	-.1212785	.7805059
tau_u	.3590843	.2271338	1.58	0.114	-.0860898	.8042584

Additional estimators with kappalate

```
. kappalate lwage (somecol=nearc4) black south smsa smsa66 region exper expers  
> q, zmodel(logit) which(all)
```

Weighting estimation of the LATE

Outcome : lwage
Treatment : somecol
Instrument : nearc4
IPS : logit ML

Number of obs = 3010

lwage	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
tau_a	1.25635	.8277096	1.52	0.129	-.3659313	2.878631
tau_a,1	1.202343	.7589756	1.58	0.113	-.2852216	2.689908
tau_a,0	1.395732	1.062338	1.31	0.189	-.6864125	3.477876
tau_a,10	.3296137	.2300512	1.43	0.152	-.1212785	.7805059
tau_u	.3590843	.2271338	1.58	0.114	-.0860898	.8042584

Covariate balancing with lateffects

```
. lateffects balancing (lwage) (somecol) (nearc4 black south smsa smsa66 regio  
> n exper expersq)
```

Iteration 0: EE criterion = .00309879

Iteration 1: EE criterion = .00004532

Iteration 2: EE criterion = 6.977e-13

Iteration 3: EE criterion = 1.917e-25

Local average treatment effect

Number of obs = 3,010

Estimator: Normalized covariate balancing

Outcome model: Weighted mean

Treatment model: Weighted mean

IV pscore model: Logit

		Robust				
	lwage	Coefficient	std. err.	z	P> z	[95% conf. interval]
LATE						
somecol						
(Yes vs No)		.4248859	.2510863	1.69	0.091	-.0672342 .917006

Same balancing estimate with kappalate

```
. kappalate lwage (somecol=nearc4) black south smsa smsa66 region exper expers  
> q, zmodel(cbps)
```

Weighting estimation of the LATE

```
Outcome      :   lwage  
Treatment    :   somecol  
Instrument    :   nearc4  
IPS          :   logit CB
```

```
Number of obs   =   3010
```

lwage	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----						
tau_u	.4248859	.2510863	1.69	0.091	-.0672342	.917006

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drlate

Syntax: `drlate (ovar [omvarlist] [, omodel]) (tvar [tmvarlist] [, tmodel]) (iv [ivvarlist] [, ivmodel]) [if] [in] [weight] [options]`

where *omodel* and *tmodel* can be linear, logit, or poisson, and *ivmodel* can be logit, cbps, or ipt

Essential options:

- `late` vs. `latt` determines whether the LATE or LATT is estimated
- `method(string)`, where *string* may be `ipwra` (default), `ipw`, `aipw`, or `ra`, which selects the estimator to be used

drlate vs. lateffects

There is some overlap between `drlate` and `lateffects` (and between `drlate` and `kappalate`), but also some major differences.

Advantages of `drlate`:

- A broader set of estimators, including regression adjustment (RA) and augmented IPW (AIPW)
- Allows inverse probability tilting (IPT) estimation of the instrument propensity score
- Allows estimation of the LATE or LATT
- Separately displays estimates and standard errors for the reduced form and first stage coefficients
- Like `kappalate`, it allows nonbinary treatments, automatically standardizes nonbinary covariates, and can be used in Stata 17, 18, and 19
- Like `lateffects`, it works with weighted data

Advantages of `lateffects`:

- As before; in addition, it supports additional outcome (fractional logit, fractional probit, probit) and treatment (probit) models

IPWRA with lateffects

```
. lateffects ipwra (lwage black south smsa smsa66 region exper expersq) (somec  
> ol black south smsa smsa66 region exper expersq) (nearc4 black south smsa sm  
> sa66 region exper expersq)
```

Iteration 0: EE criterion = 1.307e-18

Iteration 1: EE criterion = 6.266e-30

Local average treatment effect

Number of obs = 3,010

Estimator: IPW regression adjustment

Outcome model: Linear

Treatment model: Logit

IV pscore model: Logit

		Robust				
	lwage	Coefficient	std. err.	z	P> z	[95% conf. interval]
LATE						
somecol						
(Yes vs No)		.3868129	.2386614	1.62	0.105	-.0809549 .8545806

IPWRA with drlate

```
. drlate (lwage black south smsa smsa66 region exper expersq) (somecol black s  
> outh smsa smsa66 region exper expersq) (nearc4 black south smsa smsa66 regio  
> n exper expersq)
```

Local average treatment effect Number of obs = 3,010

Estimator : IPWRA

Outcome model : linear

Treatment model : logit

Instrument model : logit (MLE)

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
LATE							
	D on Y	.3868129	.2388484	1.62	0.105	-.0813214	.8549471
ATE							
	Z on Y	.0307378	.0189267	1.62	0.104	-.0063578	.0678334
	Z on D	.0794643	.0187635	4.24	0.000	.0426885	.1162401

IPWRA with inverse probability tilting

```
. drlate (lwage black south smsa smsa66 region exper expersq) (somecol black s  
> outh smsa smsa66 region exper expersq) (nearc4 black south smsa smsa66 regio  
> n exper expersq, ipt)
```

```
Local average treatment effect          Number of obs   =       3,010  
Estimator           : IPWRA  
Outcome model       : linear  
Treatment model      : logit  
Instrument model     : logit (IPT)
```

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
LATE							
	D on Y	.3848827	.2388961	1.61	0.107	-.0833451	.8531105
ATE							
	Z on Y	.0305789	.0189166	1.62	0.106	-.006497	.0676548
	Z on D	.07945	.0187977	4.23	0.000	.0426072	.1162928

LATT estimation with drlate

```
. drlate (lwage black south smsa smsa66 region exper expersq) (somecol black s  
> outh smsa smsa66 region exper expersq) (nearc4 black south smsa smsa66 regio  
> n exper expersq), latt
```

Local average treatment effect on the treated Number of obs = 3,010
Estimator : IPWRA
Outcome model : linear
Treatment model : logit
Instrument model : logit (MLE)

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
LATT							
	D on Y	.2999445	.2365009	1.27	0.205	-.1635889	.7634778
ATT							
	Z on Y	.0276518	.0219331	1.26	0.207	-.0153362	.0706399
	Z on D	.0921898	.0217054	4.25	0.000	.0496479	.1347316

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teffects2

teffects2 is designed to resemble teffects in syntax and use, with sub-commands `aipw`, `ipw`, and `ipwra`. Unlike teffects, however, **it allows the treatment propensity score to be estimated using IPT or CBPS**.

Both approaches **balance covariates** between the relevant subsamples:

- **Inverse probability tilting (IPT)**, as in Graham et al. (2012, 2016), has attractive theoretical properties, but convergence can sometimes be challenging
- The **covariate balancing propensity score (CBPS)** approach of Imai and Ratkovic (2014) is perhaps more widely known
- IPT and CBPS differ for the ATE, but are the same (and also the same as Hainmueller's "entropy balancing") when estimating the ATT
- Reminder: Using IPT implies that the IPW, AIPW, and IPWRA estimators are numerically identical when the augmentation or regression adjustment models are linear

IPW with teffects2

```
. quietly webuse cattaneo2  
  
. generate mage2 = mage^2  
  
. teffects2 ipw (bweight) (mbsmoke mmarried mage mage2 fbaby medu, logit)
```

Treatment effect estimation Number of obs = 4,642
Estimator : IPW
Outcome model : weighted mean
Treatment model: logit ML

		Robust				
bweight	Coefficient	std. err.	z	P> z	[95% conf. interval]	
ATE	-231.7203	25.17974	-9.20	0.000	-281.0717	-182.3689
POmean	3403.527	9.576358	355.41	0.000	3384.757	3422.296

Same IPW estimates with teffects

```
. teffects ipw (bweight) (mbsmoke mmarrried mage mage2 fbaby medu, logit)
```

Iteration 0: EE criterion = 1.714e-21

Iteration 1: EE criterion = 6.041e-27

Treatment-effects estimation	Number of obs	=	4,642
------------------------------	---------------	---	-------

Estimator : inverse-probability weights

Outcome model : weighted mean

Treatment model: logit

			Robust				
	bweight	Coefficient	std. err.	z	P> z	[95% conf. interval]	
ATE							
	mbsmoke						
	(Smoker						
	vs						
	Non smoker)	-231.7203	25.17975	-9.20	0.000	-281.0717	-182.3689
POmean							
	mbsmoke						
	Non smoker	3403.527	9.576358	355.41	0.000	3384.757	3422.296

IPW with CBPS

```
. teffects2 ipw (bweight) (mbsmoke mmarried mage mage2 fbaby medu, cbps)
```

```
Treatment effect estimation      Number of obs      =      4,642
Estimator      : IPW
Outcome model   : weighted mean
Treatment model: logit CBPS
```

		Robust					
bweight		Coefficient	std. err.	z	P> z	[95% conf. interval]	
-----+-----							
ATE		-229.4702	25.9892	-8.83	0.000	-280.4081	-178.5323
P0mean		3402.379	9.761902	348.54	0.000	3383.246	3421.512

IPW with inverse probability tilting

```
. teffects2 ipw (bweight) (mbsmoke mmarried mage mage2 fbaby medu, ipt)
```

```
Treatment effect estimation          Number of obs      =          4,642
Estimator       : IPW
Outcome model   : weighted mean
Treatment model : logit IPT
```

		Robust				
bweight	Coefficient	std. err.	z	P> z	[95% conf. interval]	
ATE	-225.7973	27.30981	-8.27	0.000	-279.3236	-172.2711
P0mean	3403.96	9.520665	357.53	0.000	3385.3	3422.62

IPWRA equals IPW under IPT

```
. teffects2 ipwra (bweight mmarried mage mage2 fbaby medu) (mbsmoke mmarried m
> age mage2 fbaby medu, ipt)
```

Treatment effect estimation	Number of obs	=	4,642
Estimator	: IPW regression adjustment		
Outcome model	: linear		
Treatment model	: logit IPT		

		Robust				
bweight	Coefficient	std. err.	z	P> z	[95% conf. interval]	
ATE	-225.7973	27.30981	-8.27	0.000	-279.3236	-172.2711
P0mean	3403.96	9.520665	357.53	0.000	3385.3	3422.62

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Summary

The three commands complement `teffects` and `lateffects`:

- `kappalate` provides a broader set of weighting estimators of the LATE, with the instrument propensity score estimated by maximum likelihood or covariate balancing
- `drlate` provides doubly robust IPWRA estimation of the LATE and LATT, together with IPW, AIPW, and RA alternatives
- `teffects2` follows the familiar syntax of `teffects` while adding IPT and CBPS estimation of the treatment propensity score
- With IPT and linear outcome models, IPW, AIPW, and IPWRA estimators of the ATE and ATT are numerically identical

All three packages are available from SSC.